

Speaker Recognition Project Report

Method

This project implements a text-dependent speaker recognition system using the phrase "My voice is my passport. Verify me." For each speaker, recordings 1 through 3 were used to build a training model, and recordings 4 through 6 were used as test files. Each audio file was processed by extracting Mel-Frequency Cepstral Coefficients (MFCCs) — a compact representation of the spectral shape of the voice — using a 25ms Hamming window and 13 coefficients per frame. The mean MFCC vector across all frames was computed to serve as the speaker's fingerprint and saved as a .mat file. To identify an unknown file, its mean MFCC vector was compared to every speaker model using Euclidean distance, and the speaker with the smallest distance was selected as the predicted identity. All scores and predictions were logged in a spreadsheet.

Results

The method was tested across four datasets totaling 20 speakers and 60 test files.

Dataset	Speakers	Test Files	Accuracy
DSP_Assignment (Alex, Ben, Jacob, Maddie, Matt)	5	15	93.3% (14/15)
Recordings (Akash, Daniel, Eliza, Google, Steven)	5	15	86.7% (13/15)
Dataset3 (Brian, comp, grant, jay, sara)	5	15	100.0% (15/15)
speakerIDdataset7 (BrandiC, DavidL, DavidR, ErinL, WillR)	5	15	93.3% (14/15)
OVERALL	20	60	93.3% (56/60)

The system achieved an overall accuracy of 93.3% (56/60), with one dataset reaching perfect accuracy. The four errors all occurred between acoustically similar speaker pairs with very small score margins, suggesting the mistakes were not random.

Patterns and Observations

Examining the spreadsheets revealed several clear patterns. First, errors were not random — they consistently occurred between specific speaker pairs. Matt and Alex (DSP_Assignment) had the closest scores to each other across every test file, and their one confusion (Matt_5 predicted as Alex) reflects a genuine acoustic similarity between the two voices. Similarly, Daniel and Eliza (Recordings) confused each other in both directions, suggesting their voices have similar spectral characteristics. DavidR and WillR produced one confusion with a very small margin of difference.

A second notable pattern was that synthetic or computer-generated voices were identified perfectly and with large margins in two datasets. The speaker labeled "Google" had self-scores below 0.85 but scores against humans of 4.2 to 6.0, and the speaker labeled "comp" showed a similar pattern with even larger margins (up to 10.3

against some speakers). This makes intuitive sense — synthesized voices have a highly uniform spectral profile that is distinctly different from any natural human voice.

Overall, speakers whose scores were consistently well-separated from all others were identified reliably every time, while speakers whose scores were close to another specific speaker were the only source of errors. This suggests the method is sound but limited by cases of genuine vocal similarity.

Challenges

The main technical challenge was a syntax difference in MATLAB's `mfcc()` function — the `WindowLength` parameter is not supported in all versions and had to be replaced with the `Window` parameter, which requires passing an explicit Hamming window vector. A second challenge was that the four datasets used different file naming conventions, requiring small adjustments to the file-loading code for each dataset. More broadly, using a simple mean MFCC template as the speaker model discards temporal information about each speaker's voice. A Gaussian Mixture Model (GMM) would likely improve accuracy by capturing the full distribution of each speaker's MFCC frames rather than just the average.